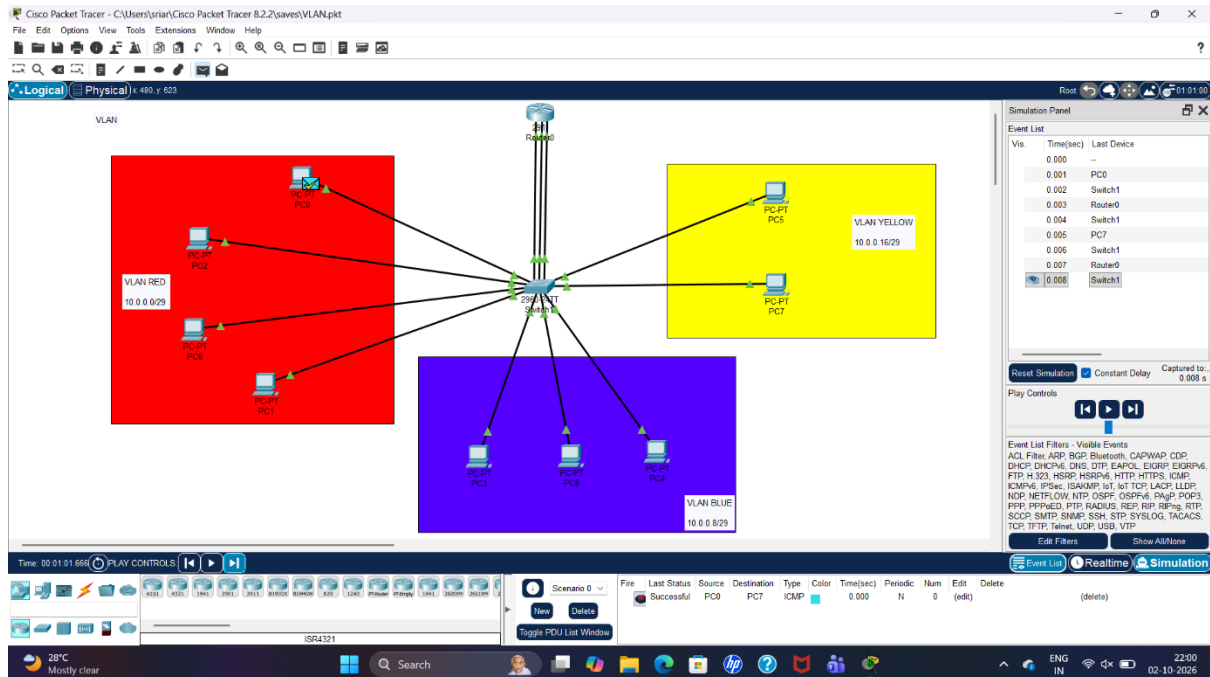


Project title: VLAN.

Objective: To design and configure a network using three separate VLANs, each associated with a different subnet and connected through a dedicated router link.

Network topology:



Verification:

Device: switch.

Command: show vlan brief

```
Switch>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	RED	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/10
11	blue	active	Fa0/5, Fa0/6, Fa0/7, Fa0/11
12	yellow	active	Fa0/8, Fa0/9, Fa0/12
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch>
```

Device: Router.

Command: Show ip route

```

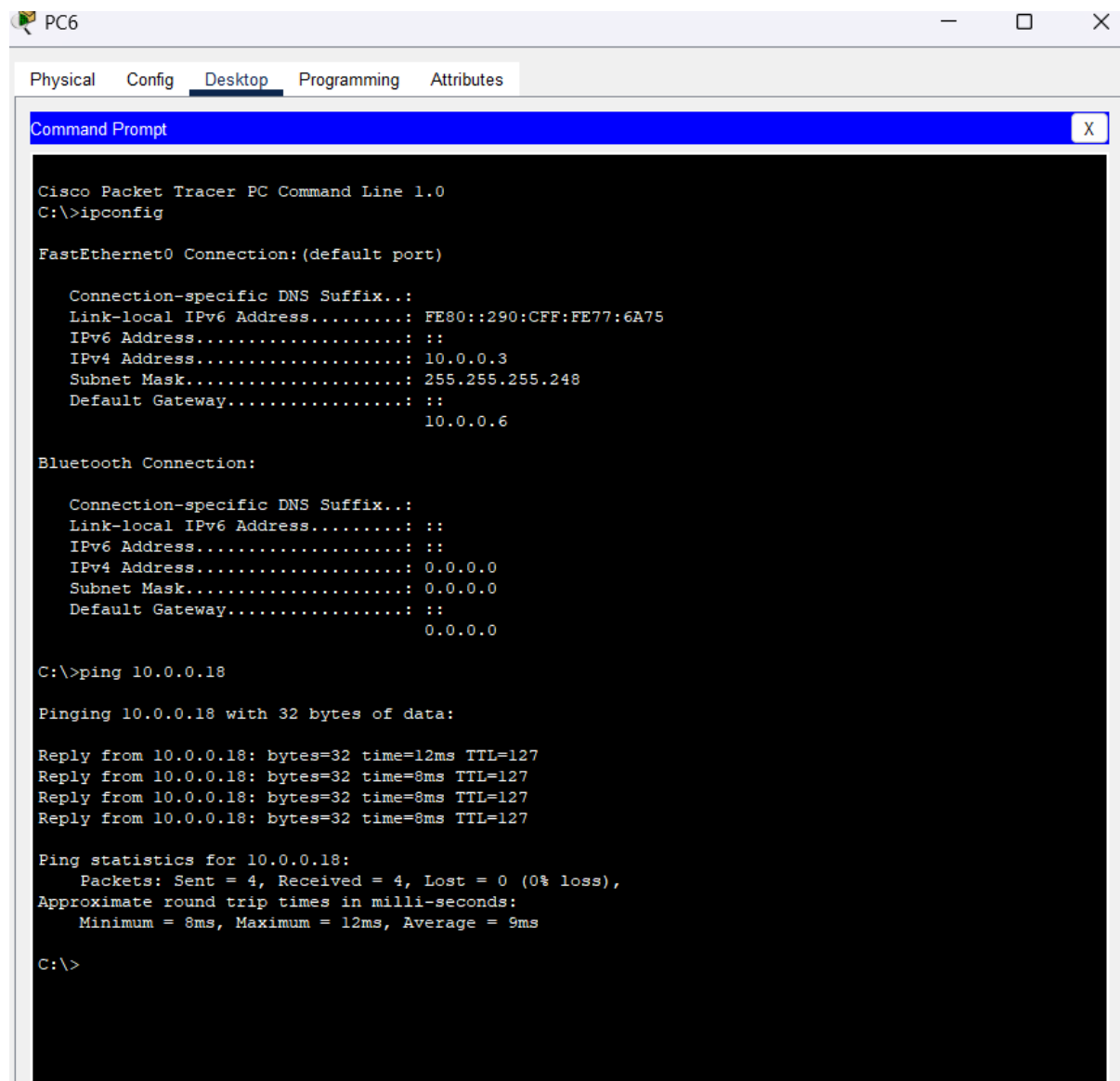
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.0.0/29 is directly connected, GigabitEthernet0/0
L       10.0.0.6/32 is directly connected, GigabitEthernet0/0
C       10.0.0.8/29 is directly connected, GigabitEthernet0/1
L       10.0.0.14/32 is directly connected, GigabitEthernet0/1
C       10.0.0.16/29 is directly connected, GigabitEthernet0/2
L       10.0.0.22/32 is directly connected, GigabitEthernet0/2

```

Ping command from PC in VLAN RED to PC in VLAN YELLOW.



```

PC6
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::290:CFF:FE77:6A75
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 10.0.0.3
    Subnet Mask . . . . .: 255.255.255.248
    Default Gateway . . . . .: ::
                                10.0.0.6

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

C:\>ping 10.0.0.18

Pinging 10.0.0.18 with 32 bytes of data:

Reply from 10.0.0.18: bytes=32 time=12ms TTL=127
Reply from 10.0.0.18: bytes=32 time=8ms TTL=127
Reply from 10.0.0.18: bytes=32 time=8ms TTL=127
Reply from 10.0.0.18: bytes=32 time=8ms TTL=127

Ping statistics for 10.0.0.18:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 12ms, Average = 9ms

C:\>

```